

Full budget permits zero oracle queries

A counterexample to Conjecture 00000007863's linear lower bound

The source claim. The bilingual source asserts that the expected query complexity $q(n, k)$ of randomized value-oracle maximization of a normalized monotone submodular function is $\Theta(n)$ for a $(1 - 1/e)$ -approximation, with exact lower-bound constant one and every point queried at least once. It imposes no exclusion of $k = n$, no fixed- k asymptotic, and no restriction separating k from n . We disprove the necessary positive linear lower-bound conjunct on the unbounded admissible family $k = n$.

Every objective is handled. Let $V = \{0, \dots, n - 1\}$. The allowed objective is an actual set function $f : 2^V \rightarrow \mathbb{R}$ satisfying

$$f(\emptyset) = 0, \quad A \subseteq B \implies f(A) \leq f(B), \quad f(A \cup B) + f(A \cap B) \leq f(A) + f(B).$$

At budget $k = n$, the ground set itself is feasible. Every feasible set S is a subset of V , so monotonicity gives

$$f(S) \leq f(V).$$

Thus V attains the actual maximum, for *every* objective in the source class. This does not select an easy instance such as the zero function. The cardinality objective $f(S) = |S|$, for example, is normalized, monotone and modular, and has positive optimum n when $n > 0$. It witnesses that the class and counterexample are nonvacuous.

Approximation guarantee. Normalization and monotonicity imply $0 = f(\emptyset) \leq f(S)$ for every set. Put $\alpha = 1 - 1/\exp(1)$, so $0 < \alpha \leq 1$. Consequently

$$\alpha f(S) \leq f(S) \leq f(V).$$

Returning V therefore satisfies the requested approximation, indeed exact optimality, uniformly over the whole class. Submodularity is retained in the hypotheses, even though monotonicity alone makes this full-budget argument work.

Actual zero-query protocol. The algorithm is simply

return V .

It does not use an oracle answer or depend on which objective is present. In the formal model, a protocol is built from return nodes and query nodes; a query requests $f(S)$, feeds that real answer to its continuation, and increases the interpreter's query count by one. The return node evaluates to output V and count zero for every oracle. Zero cost is proved from this actual interpreter, not supplied as a premise or assigned to an unrelated complexity scalar.

Expected cost, asymptotics, and precise scope

A legitimate randomized special case. A deterministic protocol is allowed among randomized algorithms. Explicitly, take a singleton random seed with probability mass one. The seed law is nonnegative and normalized. Its expected query count is the finite weighted sum of the actual interpreter counts. For the same return-ground-set protocol this gives

$$\mathbb{E}[\text{queries} \mid f] = 1 \cdot 0 = 0 \quad \text{for every allowed } f.$$

Therefore even the worst case over all source objectives has zero expected queries. Query counts are nonnegative, so this uniformly valid algorithm already rules out any positive lower bound on the least required worst-case expected count at full budget.

Asymptotic contradiction. Suppose a positive linear lower bound held eventually along $(n, k) = (n, n)$. There would be $c > 0$ and a threshold such that every valid randomized approximation algorithm, for every sufficiently large n , had expected cost at least cn on some allowed objective. Apply that claim to the return-ground-set protocol. Its expected cost is zero on every objective, whereas $cn > 0$ once $n \geq 1$. This is a contradiction. Thus no eventual positive $\Omega(n)$ lower comparison is possible on this unbounded family, and the stated unrestricted $\Theta(n)$ law, including its exact-constant assertion, fails.

Scope of the conclusion. We use the standard optimization output: a feasible set with the promised objective guarantee. The source does not request separately reporting its unknown objective value. It also does not restrict the budget to $k < n$ or fix k as n grows. Those restrictions would describe a different claim. Independent source-bound review must assess these conventions.

The source additionally asserts a query jump for improved approximation and a uniquely determined information-theoretic exponent. Those clauses are not individually formalized or settled here. A false necessary linear lower-bound conjunct suffices to refute the original conjunction on its stated domain. The simple protocol class is used only to give an allowed counterexample; it does not purport to rebuild the full randomized oracle model.

Lean correspondence. `SourceObjective` retains all three source properties on actual `Finset (Fin n)` objects. The complete source proves greatest-value optimality, a nontrivial cardinality witness, and the exact approximation for every objective. `ValueProgram` and `run` implement actual oracle queries and their cost; the normalized seed expectation is proved zero. The expected-query lower-bound theorem supplies the asymptotic contradiction. The premise-free theorem `full_budget_counterexample` collects the full certificate.

Provenance and verification. This elementary full-budget route was identified by the root coordinator and formalized in the same-run scout before this author package. No first-discovery claim is made. The packet is an author artifact; deterministic replay, independent semantic review and acceptance belong to the manager.